XENON INTERPRETABILITY

Synthetic Market Representation Controls

Representation-focused checkpoint after Phase 4 rank-context probing and the Phase 5 symbol-permutation control. Research anchors: `RANKED_RESEARCH_ROADMAP.md`, `MARKET_COUNTING_MANIFOLDS_PLAN.md`, and `MARKET_REPRESENTATION_CONTROLS_NOTES.md`.

DATE	PHASE 4	PHASE 5	LAYERS
21 March 2026	69 market-only prompts	12 permuted prompts	48

MAIN READ

Primitive market variables are real; profile-level abstraction is selective.

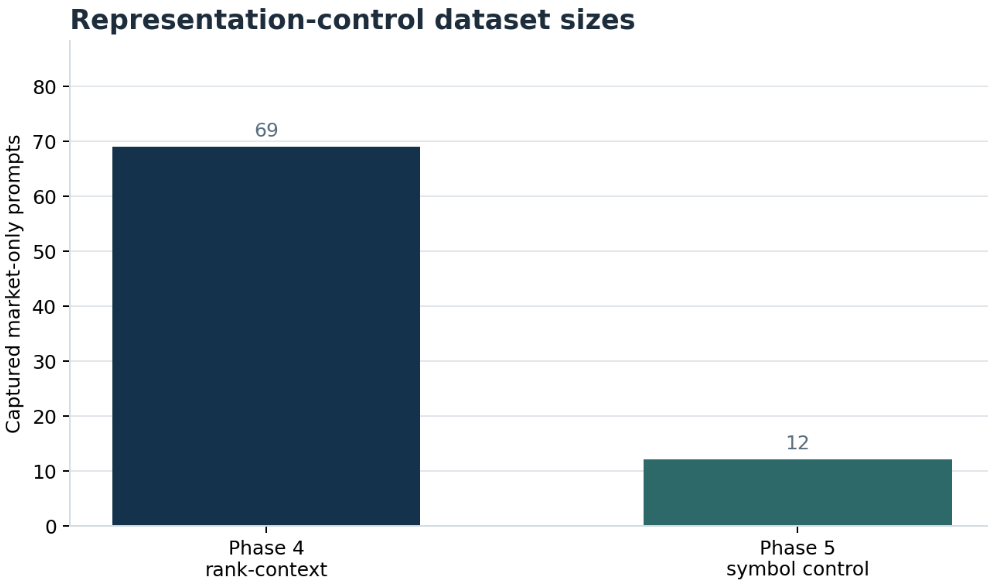
The row states preserve momentum, flow, participation, concentration, and synthetic latent scores with near-perfect linear fidelity. But once row position and display symbol shortcuts are explicitly removed, only some profile families remain clearly separable. Participation/concentration survives the hard control; momentum/flow mostly collapses.

BEST PRIMITIVE R^2	PHASE 4 CONFOUND	BEST CONTROL MARGIN	CONTROL BASELINE
0.99999	**0.523**	**0.262**	**0.188**
concentration, Phase 5	same-symbol margin	participation/concentration	random profile retrieval

Scope

This report narrows the question to market representation itself.

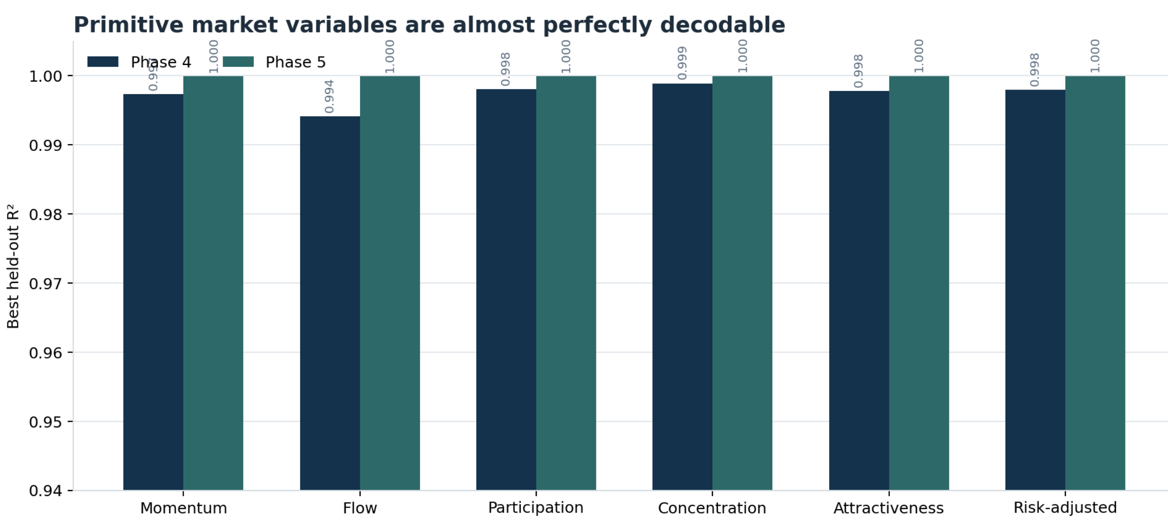
- **Not action labels.** The target is how the model encodes the market rows, not what final tool call it emits.
- **Primitive variables first.** The first test asks whether the model linearly preserves market factors like momentum, flow, participation, and concentration.
- **Confound controls next.** The second test asks whether those representations survive symbol and row permutations.
- **Synthetic-first.** Everything here is backed by synthetic datasets before any return to real DX validation.



Phase 4 and Phase 5 prompt counts. Phase 4 is the broader market-representation slice; Phase 5 is the tighter symbol-permutation control.

Primitive Factor Results

The stable result across both phases is that primitive market variables are extremely decodable from the row states.



Best held-out regression performance for primitive market variables. Both phases show near-perfect linear recovery, which means the row states preserve these variables very explicitly.

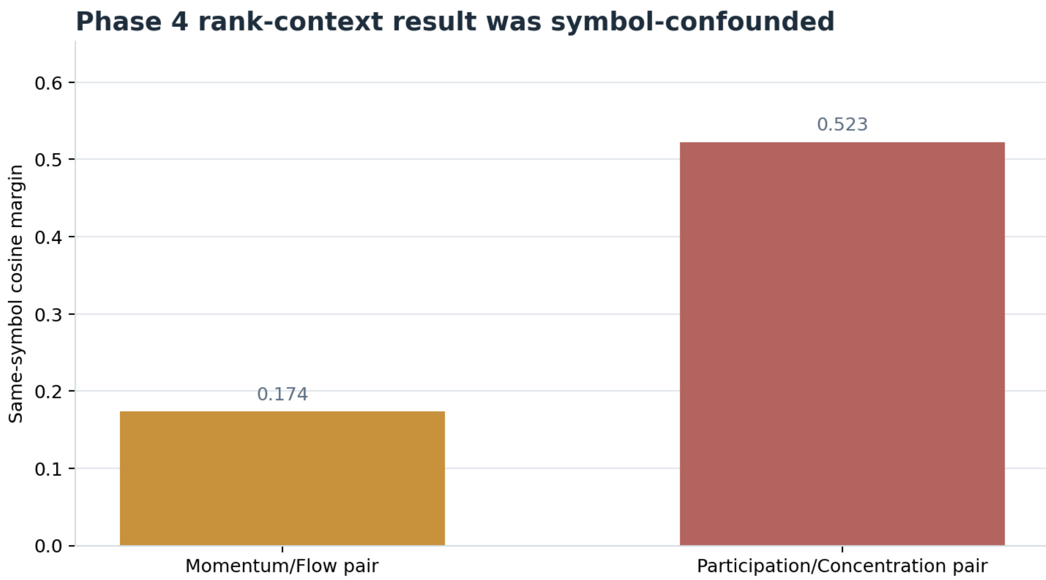
Question	Best State	Metric	Layer
`pct_5m`	`row_mean`	R^2 0.997 / 1.000	L6 / L1
`net_flow_5m`	`row_mean`	R^2 0.994 / 1.000	L6 / L1
`unique_traders_5m`	`row_mean`	R^2 0.998 / 1.000	L14 / L1

Question	Best State	Metric	Layer
`top20_holder_pct`	`row_mean`	R^2 0.999 / 1.000	L6 / L1
`attractiveness_score`	`row_mean`	R^2 0.998 / 1.000	L6 / L1
`risk_adjusted_score`	`row_mean`	R^2 0.998 / 1.000	L6 / L1

This is the strongest supported claim in the current program: the model really is carrying specific market factors in the row states. That is a representation result, not just a behavior result.

Why Phase 4 Was Not Enough

Phase 4 also included a rank-context retrieval idea: keep the focal asset pair numerically fixed while changing the background roster, then ask whether the same latent asset remains nearest to itself across variants.



Phase 4 looked strong if read naively, but these margins were still heavily driven by symbol identity and row position.

- `fixed_momentum_flow_pair`: same-symbol cosine margin `0.174`
- `fixed_participation_concentration_pair`: same-symbol cosine margin `0.523`
- Both scenarios had same-symbol nearest-neighbor accuracy `1.0`

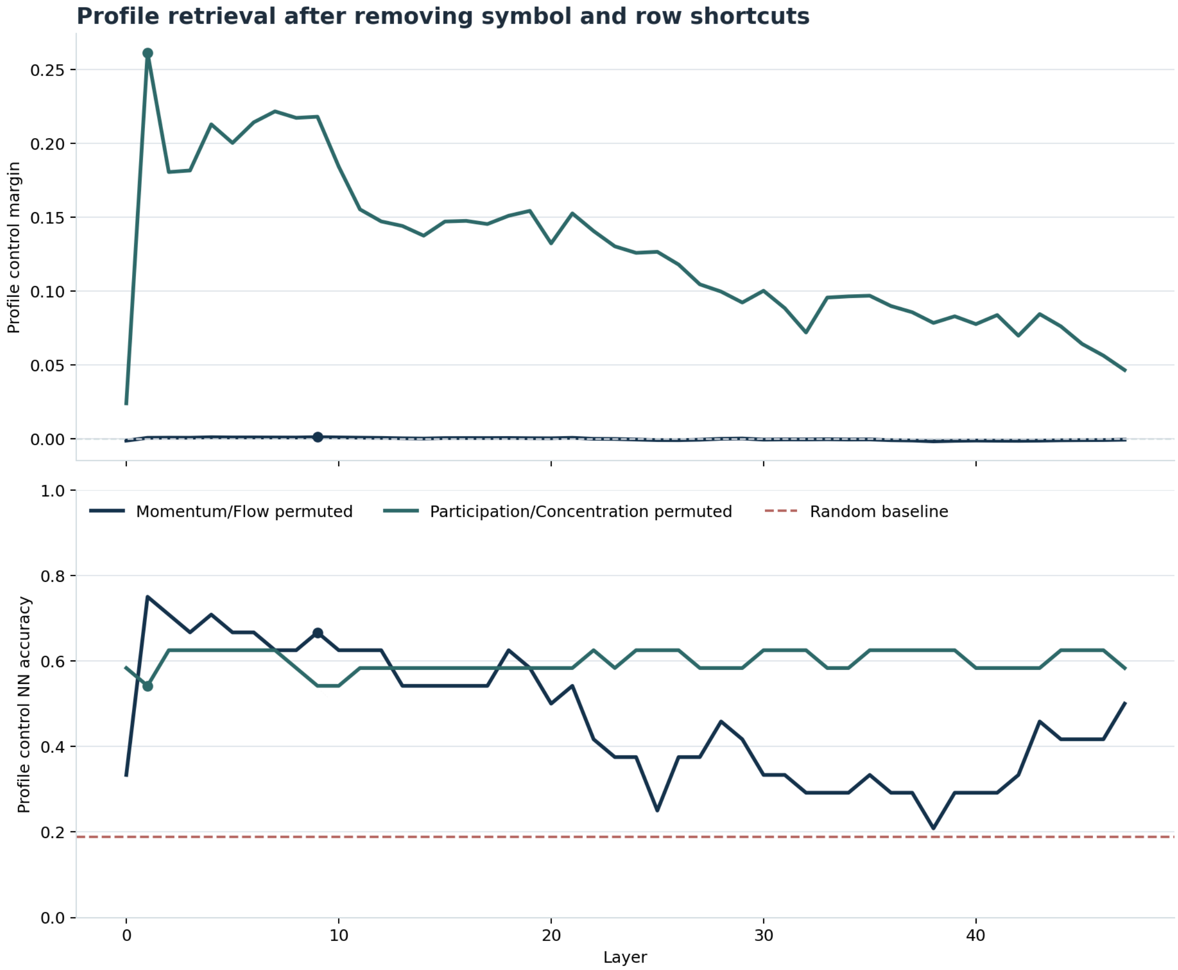
That is exactly why Phase 5 was necessary. Phase 4 showed persistence, but not whether persistence came from latent market profile or from lexical/positional shortcuts.

Phase 5 Symbol-Permutation Control

Phase 5 permutes the same latent profiles across both display symbol and row index. The main metric is stricter than the earlier nearest-neighbor read:

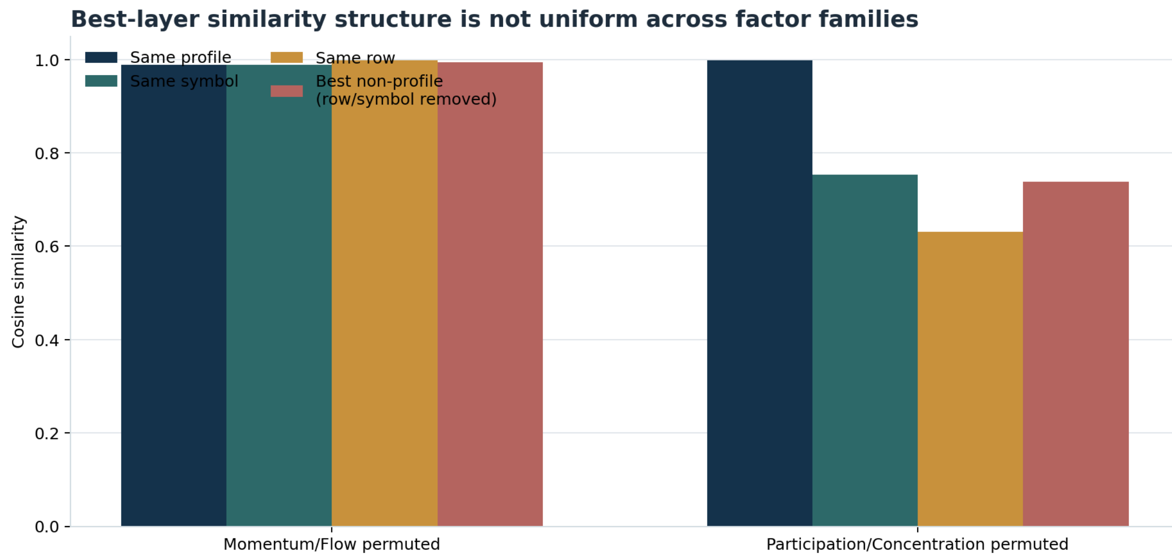
- `profile_control_nn_accuracy`
- `profile_control_margin`

These metrics exclude every candidate that shares the same display symbol or same row position, so the remaining retrieval problem is closer to the latent question we actually care about.



Layerwise control metrics after removing row and symbol shortcuts. The dashed line is the random baseline induced by the phase-5 permutation layouts (`0.188`).

What Survives The Hard Control



Best-layer similarity breakdown for each Phase 5 scenario. Participation/concentration remains profile-dominant; momentum/flow becomes nearly tied once row and symbol are removed.

Scenario	Best State	Control Margin	Control NN Acc.	Read
`momentum_flow_permuted_market`	`row_mean` L9	0.001	0.667	Above-chance retrieval exists, but cosine separation is almost fully collapsed.
`participation_concentration_permuted_market`	`raw_eos` L1	0.262	0.542	Profile identity survives the hard control meaningfully better.

At the best participation/concentration layer:

- same-profile cosine mean: `0.999`
- same-symbol cosine mean: `0.753`
- same-row cosine mean: `0.631`
- best non-profile cosine after row/symbol removal: `0.738`

At the best momentum/flow layer:

- same-profile cosine mean: `0.989`
- same-symbol cosine mean: `0.988`
- same-row cosine mean: `0.998`
- best non-profile cosine after row/symbol removal: `0.994`

So the participation/concentration family looks like a genuine profile-level abstraction candidate. The momentum/flow family does not, at least not yet.

Why The AUROC 1.00 Results Are Not The Headline

> **CAUTION**
>
> `AUROC 1.000` on the hard pairwise synthetic slices does not mean we found a rich market semantics circuit. It mainly shows that the primitive synthetic tradeoff labels are linearly recoverable in an intentionally clean setting.

This is why the report now emphasizes control-based profile retrieval rather than pairwise classification:

- pairwise AUROC can be perfect while still reflecting a very easy synthetic decision boundary
- confound controls tell us more about abstraction
- the real question is not whether the model can recover synthetic labels, but whether the same latent asset profile remains stable after superficial prompt identity changes

Interpretation

> **SUPPORTED NOW**
>
> Primitive market variables are explicitly present in row states, and at least one profile family survives a hard row/symbol permutation control.

> **NOT SUPPORTED YET**
>
> We do not yet have evidence for a single, uniformly robust profile-level market representation across all factor families.

Best current interpretation:

- The model has strong early access to primitive market factors.
- Some higher-order abstractions exist, but they are selective.
- Participation/concentration looks substantially more stable under symbolic relabeling than momentum/flow.
- The right next target is not a global market manifold claim; it is controlled, family-specific abstraction testing.

Next Step

The most defensible next experiment is a harder synthetic profile-invariance pass:

1. preserve the Phase 5 permutation control
2. add row-text paraphrases and order perturbations
3. create near-tied momentum/flow cases with sharper participation/concentration differences
4. rerun the same profile-control metrics before bringing the strongest surviving family back to real DX rows

If participation/concentration remains stable under those stronger wording controls, that becomes the best candidate for real-data validation.
